SM90 FP4 MegaMoE | NVIDIA H20-3e

82 workload configurations | NVFP4 fused, NVFP4 split, MXFP4 fused | current kernel latency

Cold L2 before each call; CUDA events inside a graph; per-call max across ranks, then median of 50 samples.

NVFP4 | fused + split

24 workloads / EP8

M / rank	Capacity	Kernel latency (us)
NVFP4 FUSED | Flash EP8		
1	384	**90.960**
24	384	**254.016**
32	384	**258.624**
64	384	**288.208**
96	384	**327.056**
NVFP4 FUSED | Mimo EP8		
1	384	**151.376**
24	384	**509.616**
32	384	**519.904**
64	384	**579.584**
96	384	**629.312**
NVFP4 FUSED | Pro EP8		
1	384	**169.152**
24	384	**849.856**
32	384	**890.464**
64	384	**930.048**
96	384	**972.128**
NVFP4 SPLIT | Flash EP8		
384	4224	**891.280**
1024	4224	**1,588.000**
4096	4224	**5,223.680**
NVFP4 SPLIT | Mimo EP8		
384	4224	**1,637.664**
1024	4224	**3,027.472**
4096	4224	**10,236.048**
NVFP4 SPLIT | Pro EP8		
384	4224	**2,026.080**
1024	4224	**4,358.480**
4096	4224	**13,342.016**

MXFP4 | fused

24 workloads / full token range

M / rank	Capacity	Kernel latency (us)
MXFP4 FUSED | Flash EP8		
1	4224	**98.144**
24	4224	**271.648**
32	4224	**270.912**
64	4224	**289.104**
96	4224	**310.480**
384	4224	**801.184**
1024	4224	**1,504.720**
4096	4224	**5,195.056**
MXFP4 FUSED | Mimo EP8		
1	4224	**131.008**
24	4224	**501.568**
32	4224	**561.024**
64	4224	**582.112**
96	4224	**626.848**
384	4224	**1,436.624**
1024	4224	**2,857.840**
4096	4224	**10,136.752**
MXFP4 FUSED | Pro EP8		
1	4224	**221.840**
24	4224	**787.984**
32	4224	**836.160**
64	4224	**935.568**
96	4224	**975.072**
384	4224	**1,708.752**
1024	4224	**4,049.184**
4096	4224	**13,173.744**

MXFP4 | fused

34 workloads / EP8 + EP4

M / rank	Capacity	Kernel latency (us)
MXFP4 FUSED | Flash EP8		
8	384	**214.080**
16	384	**238.896**
32	384	**271.248**
48	384	**280.800**
64	384	**289.040**
96	384	**307.504**
112	384	**318.624**
128	384	**351.120**
160	384	**438.320**
192	384	**440.304**
256	384	**457.088**
384	384	**814.176**
MXFP4 FUSED | Flash EP4		
8	768	**261.088**
16	768	**380.096**
32	768	**438.016**
48	768	**451.552**
64	768	**496.848**
96	768	**518.560**
128	768	**535.520**
160	768	**555.216**
192	768	**573.584**
224	768	**589.936**
256	768	**614.192**
384	768	**857.952**
512	768	**866.240**
768	768	**1,519.440**
MXFP4 FUSED | Mimo EP4		
8	384	**527.168**
32	384	**854.064**
64	384	**1,008.064**
128	384	**1,101.408**
192	384	**1,216.400**
240	384	**1,204.288**
288	384	**1,304.784**
384	384	**1,804.144**

Times in microseconds. Capacity = allocated maximum tokens per rank. NVFP4 split timing includes both L1 and L2.

Flash: H4096 / I2048 / E256 / K6 | MiMo: H6144 / I2048 / E384 / K8 | Pro: H7168 / I3072 / E384 / K6

All measured configurations and latencies are provided in megamoe_sm90_performance.csv.